

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-Masked, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab in Participants With Moderate-to-Severe Thyroid Eye Disease **Short Title/Acronym:** SatraGO-1 **Protocol Number:** GP44467 / 2023-503309-13-00 **NCT Number:** NCT05987423 **Posting ID:** 706325762556584387 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Company Name:** Roche **Investigational Product:** Satralizumab (subcutaneous injection) **Phase of Development:** Phase III **Medical Monitor:** Roche Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of satralizumab in reducing proptosis (eye bulging) in participants with active, moderate-to-severe Thyroid Eye Disease (TED). **Secondary Objectives:** To assess the efficacy, safety, pharmacokinetics, and pharmacodynamics of satralizumab, including improvements in diplopia (double vision), reduction in Clinical Activity Score (CAS), and the absence of motility-induced pain.

2.2 Background & Rationale Thyroid eye disease (TED) is a rare, autoimmune, orbital inflammatory disorder that is disfiguring, debilitating, and potentially sight-threatening. There is a significant unmet need for a fast-acting, durable, systemic disease-modifying therapy for active TED. Satralizumab is a recombinant humanized anti-interleukin-6 receptor (IL-6R) monoclonal antibody that prevents IL-6 from binding to its receptor, thereby reducing proinflammatory and profibrotic signaling, offering a potentially safer and more effective targeted treatment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Double-Masked) **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration Target \$N\$: 131 enrolled participants **Number of Sites:** Multicenter across the EU, USA, and other countries **Estimated Study Start:** 26-Oct-2023 **Estimated Primary Completion:** 08-Jul-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years (or ≥ 19 years in Korea). 2. Clinical diagnosis of TED based on CAS. For active, moderate-to-severe TED: CAS ≥ 3 during screening and Day 1, and onset of active symptoms ≤ 12 months before Day 1 in the study eye. 3. Proptosis ≥ 3 mm above normal for participant's race and sex.

4.2 Exclusion Criteria 1. Decrease in CAS or proptosis of ≥ 2 points or ≥ 2 mm, respectively, in the study eye between Screening and Study Baseline (Day 1). 2. Requiring immediate surgical ophthalmological intervention or planning corrective surgery or irradiation during the course of the study, in the judgment of the investigator. 3. Identified pre-existing ophthalmic disease that, in the judgment of the investigator, would preclude study participation or complicate interpretation of study results, including corneal decompensation unresponsive to medical management and including ophthalmic diseases that will likely require prohibited therapy during the study. 4. Any serious medical condition or abnormality in clinical laboratory tests that, in the investigator's judgment, precludes an individual's safe participation in and completion of the study. 5. Pregnant or breastfeeding, or intention of becoming pregnant during the study or within 3 months after the final dose of satralizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Product Details: Satralizumab (Trade Name: Enspryng, ATC Code: L04AC19) or matching placebo. **Dosage/Frequency:** Injected every 2 weeks for the first 3 doses, then every 4 weeks (Q4W) for 5 months. Subsequent treatment is individualized and proptosis response-based. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 24 weeks for Treatment Part I, followed by 24 weeks for Treatment Part II (up to 72 weeks of total study duration).

5.2 Concomitant & Prohibited Therapy Permitted: Rescue intervention is permitted starting from Week 4 based on protocol-defined criteria as assessed by the investigator. **Prohibited:** Prior corticosteroid use within 1 month of screening, or previous systemic nonbiologic/biologic immunomodulatory agents for TED.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	-4 to -2 weeks	Informed Consent, Medical/Ophthalmic History, CAS/Proptosis baseline checks
Baseline	Day 1	Randomization, First SC Injection, Pharmacokinetics labs
Interim	Q4W (Every 4 weeks)	SC

injections, CAS assessment, exophthalmometry, safety labs, diplopia assessment | | **End of Study** | Week 72 | Final evaluations, Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of participants achieving ≥ 2 mm reduction in proptosis from Baseline (Day 1) at Week 24 in the study eye, provided there is no deterioration of proptosis (≥ 2 millimeters [mm] increase) in the fellow eye. **Measurement Method:** Clinical exophthalmometry assessments.

7.2 Secondary & Exploratory Endpoints 1. Change in proptosis from baseline at Week 24. 2. Percentage of participants achieving ≥ 1 grade reduction/improvement in diplopia among participants with baseline diplopia. 3. Percentage of participants achieving absence of motility-induced pain.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection via regular Q4W onsite visits and remote telephone contacts. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) and Safety Set. **Sample Size Justification:** A fixed-sequencing test approach was used to control the overall type I error rate for hypothesis testing of the primary and key secondary endpoints. **Handling of Missing Data:** Multiple imputation was utilized to handle missing data based on the Missing At Random (MAR) assumption.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring visits supplemented by remote patient follow-ups via telephone. **Audit Trail:** Efficacy assessment tracking, eCRF logging, and blinded data management.

11. Study Identifiers & Governance

Primary ID: GP44467 **Registry Number:** NCT05987423 **Posting ID:** 706325762556584387 **Sponsor/Lead Organization:** Roche **Study Title:**

SatraGO-1 **Official Regulatory Title:** A Phase III, Randomized, Double-Masked, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab in Participants With Moderate-to-Severe Thyroid Eye Disease

12. Clinical Framework (Thyroid Eye Disease Specific)

Disease Areas: Thyroid Eye Disease, TED **Primary Condition:** Thyroid Eye Disease (TED) **Diagnosis Threshold:** Proptosis ≥ 3 mm above normal for race/sex; CAS ≥ 3 **Medical Methodology:** IL-6 receptor inhibition therapy **Optimization Target:** Reduction of ≥ 2 mm in proptosis and ≥ 1 -grade improvement in diplopia

13. Study Procedures & Methodology

Management Pathway: Subcutaneous monoclonal antibody therapy for moderate-to-severe TED **Education Protocol (HCP):** Standardized ophthalmic assessment training for CAS, diplopia grading, and exophthalmometry **Education Protocol (Patient):** Site training on AE reporting and injection safety checks **Quality Audit Mechanism:** Ophthalmic assessment tracking & AE adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 72 weeks of study protocol + 6 months post-treatment follow-up **Onsite Visit Cadence:** Every 4 weeks (Q4W) **Remote Monitoring Frequency:** Telephone follow-up every 2 months for 6 months after the last dose **Checklist Review Interval:** Routine checks at each Q4W visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				